Landslide Dataset Description

Implementation paper : EEGMoE: A Domain-Decoupled Mixture-of-Experts Model for Self-Supervised EEG Representation Learning

Drive Link:
https://drive.google.com/file/d/1P3BZez5bBwFidgadsw64ei3WO8PdLX0N/view?usp=sharing

Implementing the above paper for predicting the possibility of landslides using the given Datasets. Use the landslide atlas as ground truth for training, testing and validation.

Core Objective: Participants are required to replicate the exact architecture proposed in the given EEGMoE paper and adapt it for landslide Sustainability Map and Detection. While the original paper focuses on EEG representation, your task is to implement this specific Domain-Decoupled Mixture-of-Experts architecture to predict the possibility of landslides using the datasets provided below. Use the landslide atlas as ground truth for training, testing, and validation.

LandslideAtlas :
https://drive.google.com/file/d/1phUWEeg5m2DeJry13vpoN70WZW71iDfH/view?usp=sharing

1. Satellite Data

Sentinel-1 (SAR Data)

- Sensor Type: Synthetic Aperture Radar (SAR)
- Acquisition Mode: Typically IW (Interferometric Wide Swath)
- Spatial Resolution: 10 meters
- Key Characteristics:
    - Operates in C-band microwave, enabling all-weather and day–night imaging
    - Sensitive to surface roughness and moisture variations
    - Useful for detecting terrain deformation

Sentinel-2 (Multispectral Data)

- Sensor Type: Optical multispectral imaging
- Bands: 13 spectral bands (Visible, NIR, SWIR)
- Spatial Resolution:
    - 10 m (Bands 2, 3, 4, 8)
    - 20 m (Bands 5, 6, 7, 8A, 11, 12)
    - 60 m (Bands 1, 9, 10)
- Key Characteristics:

    - Provides rich spectral information for vegetation, soil, and land cover analysis
    - Enables computation of indices such as NDVI, NDWI, and NBR
    - Supports detection of vegetation loss and exposed soil in landslide regions

2. Hydrometeorological Data

Rainfall Data

- Format: Raster (grid-based representation)
- Temporal Resolution: (Specify if daily/hourly – currently missing)
- Role:
    - Acts as a primary triggering factor for landslides
    - Enables modeling of rainfall thresholds and cumulative precipitation effects

3. Soil and Surface Condition Data

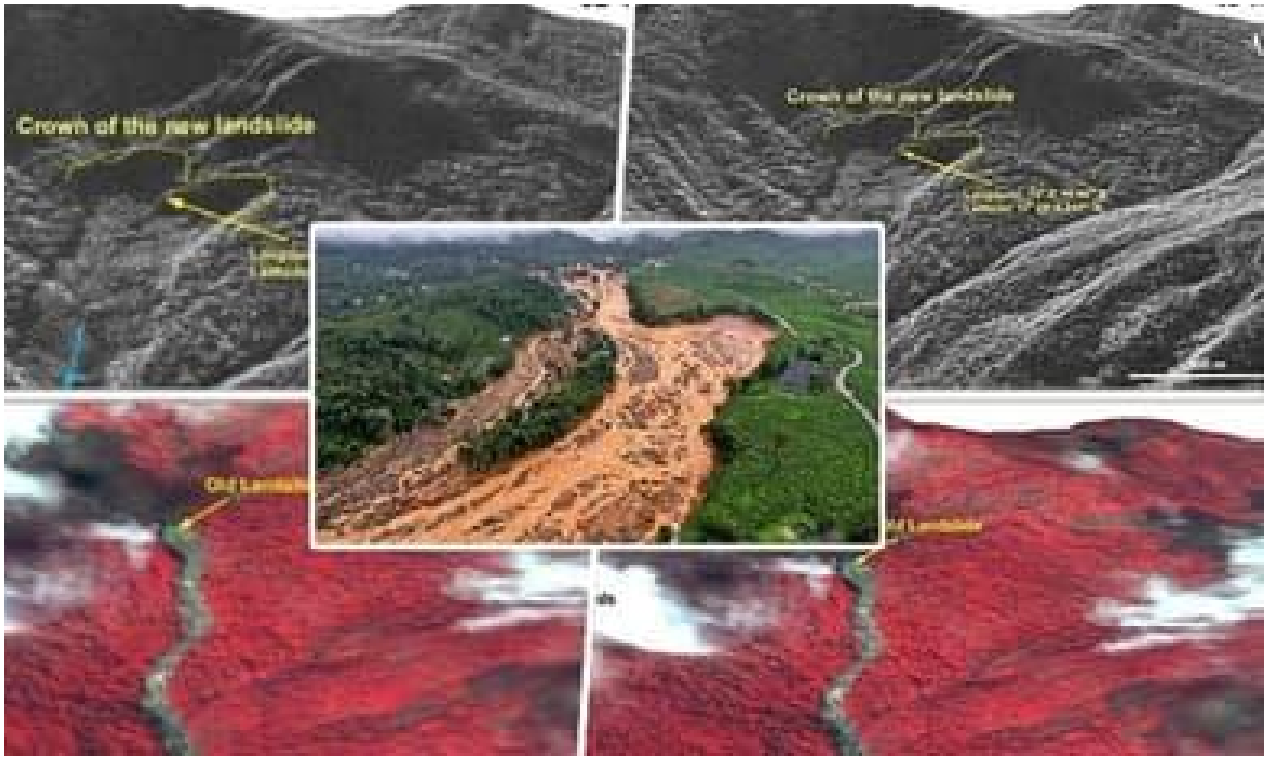
Soil Moisture Dataset

- Format: Raster data
- Source: (Specify if SMAP, ERA5, or in-situ derived – currently unclear)
- Role:
    - Indicates water content in soil layers, directly influencing slope stability
    - Critical for identifying saturated conditions leading to slope failure

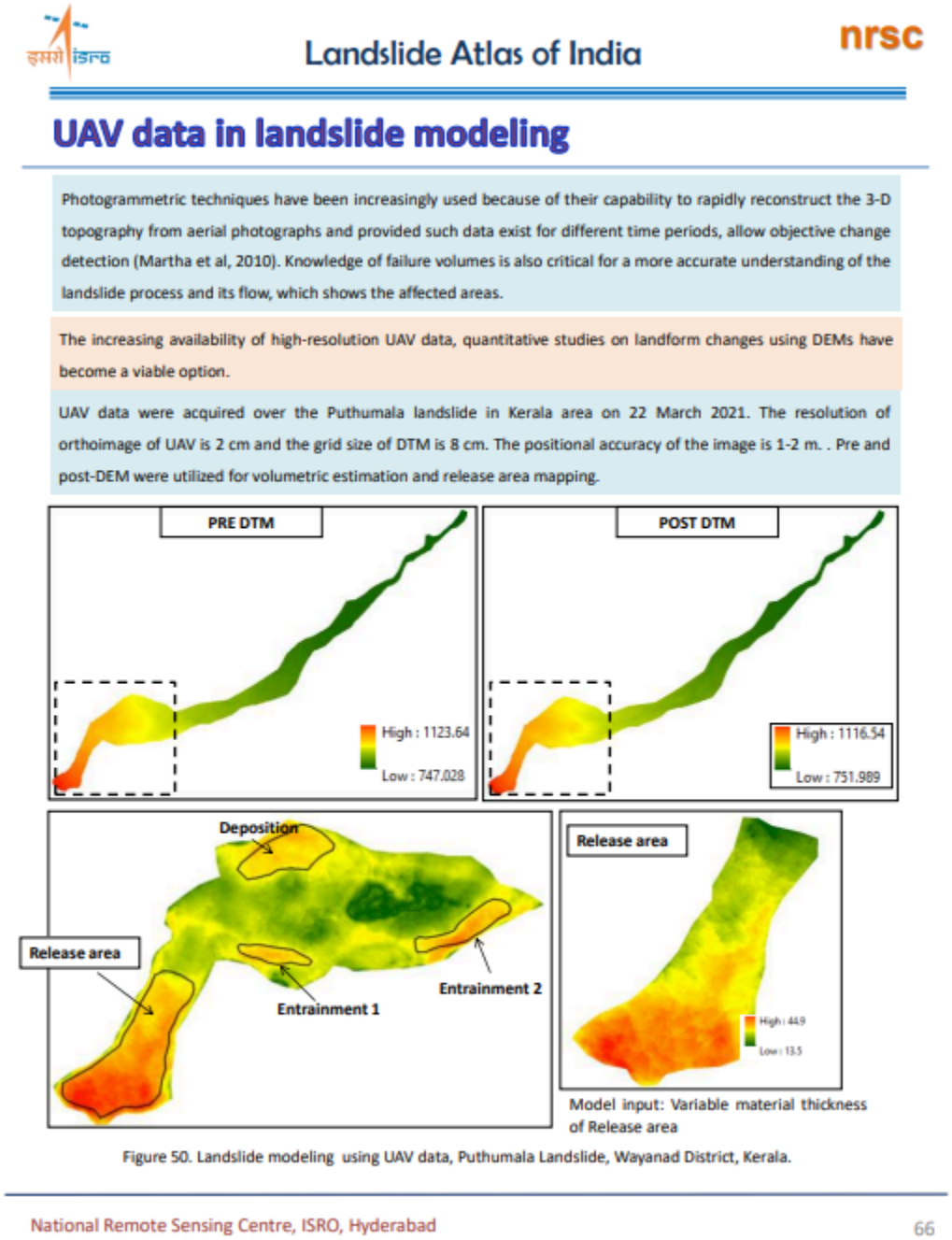
Test Cases

1. Wayanad Landslide Event (2024)

- Location: Wayanad, Kerala, India
- Event Type: Extreme rainfall-induced landslides
- Dataset Access: https://drive.google.com/drive/folders/1X42vJcGyTYTfOAzMgwKNRmGUKWyF4FJC?usp=sharing
- Reference: https://drive.google.com/drive/folders/1VZFhK6X5VefdFup_rLC9PJpERi_XQBXW?usp=sharing
- Impact Map: https://drive.google.com/file/d/1hZD3yG4Y6FeWs4fs0k9rwOGqvP_v_8gi/view?usp=sharing


Crown of the new landslide
Crown of the new landslide

2. Puthumala Landslide Event (2019)



Landslide Atlas of India

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UAV data in landslide modeling

Photogrammetric techniques have been increasingly used because of their capability to rapidly reconstruct the 3-D topography from aerial photographs and provided such data exist for different time periods, allow objective change detection (Martha et al, 2010). Knowledge of failure volumes is also critical for a more accurate understanding of the landslide process and its flow, which shows the affected areas.

The increasing availability of high-resolution UAV data, quantitative studies on landform changes using DEMs have become a viable option.

UAV data were acquired over the Puthumala landslide in Kerala area on 22 March 2021. The resolution of orthoimage of UAV is 2 cm and the grid size of DTM is 8 cm. The positional accuracy of the image is 1-2 m. . Pre and post-DEM were utilized for volumetric estimation and release area mapping.

Figure 50. Landslide modeling using UAV data, Puthumala Landslide, Wayanad District, Kerala.

National Remote Sensing Centre, ISRO, Hyderabad 66

- Location: Puthumalai (near Gudalur region, Tamil Nadu–Kerala border)
- Event Type: Catastrophic landslide
- Characteristics:
    - Occurred during 2019 monsoon extreme rainfall events

    - Associated with large-scale slope failure and debris flow
- Dataset Access:
    - Google Drive: https://drive.google.com/drive/folders/1sQnIyzPhhsewpaCmYCyGw8GvoCKd1j60?usp=sharing